

RICARDO LOPEZ

Aspiring Data Science Engineer

CONTACT

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ABOUT ME

Electronics and Communication Engineering student at SRM Institute of Science and Technology with a strong interest in IoT, Artificial Intelligence, Robotics, and Embedded Systems. Experienced in developing hardware and software-based solutions using ESP32, Arduino, Python, and Machine Learning technologies.

Actively participated in hackathons, securing multiple wins and recognitions for innovative projects. Possess strong problem-solving, teamwork, leadership, and project development skills. Passionate about exploring emerging technologies, building practical solutions, and continuously expanding technical expertise.

EDUCATION

SRMIST , Ramapuram *(2024 - PRESENT)*

Bachelor of Technology in ***Electronics and Communication with Data Science***

Pursuing a B.Tech in Electronics and Communication Engineering with a strong foundation in circuit design, signal processing, and telecommunication principles. Actively engaging in coursework and lab projects focused on emerging technologies, including IoT and embedded systems.

Montfort . Mat . Hr . Sec . School , St.Thomas Mount *(2022 - 2024)*

Higher Secondary Education

Completed Higher Secondary Education with a focus on Mathematics, Physics, and Chemistry, achieving a strong academic foundation.

LANGUAGES

- *ENGLISH*
- *TAMIL*
- *GERMAN (A1)*
- *TELUGU (BASIC)*

EXPERTISE

- *DATA CLEANING & DATA PREPARATION*
- *EXPLORATORY DATA ANALYSIS (EDA)*
- *DATA VISUALIZATION*
- *SQL & SPREADSHEET PROFICIENCY*
- *ANALYTICAL THINKING & COMMUNICATION*

- *LEADERSHIP SKILLS*
- *PUNCTUALITY*
- *COMMUNICATION*
- *PROBLEM SOLVING*

PATENTS

- ***FIELDEYE-AI DRIVEN PLANT HEALTH MONITORING SYSTEM***

- ***ADAPTIVE AI HET (HALL EFFECT THURSTERS) TESTBED***

ACHIVEMENTS

HACKATHONS :

NEXT GEN AI HACKS (THEME PRIZE WINNER)

REC - SOUTH CHENNAI CAMPUS (OCTOBER , 2026)

DATATHON WINNER (1ST PRIZE WINNER)

SRMIST - RAMAPURAM (SEPTEMBER , 2026)

PROTO-THON (THEME PRIZE WINNER)

HINDUSTAN ITS - OMR (NOVEMBER , 2025)

TECHNO-A-THON (1ST PRIZE WINNER)

ST JOSEPH'S COLLEGE OF ENGINEERING - OMR (SEPTEMBER , 2025)

PAPER PRESENTATION :

MICRONEX '25 (2ND PRIZE WINNER)

SIMATS ENGINEERING (APRIL , 2025)

PROJECTS

1. PARK-A-LOT – Smart IoT-Based Parking Management System

Developed an intelligent parking management platform providing real-time parking slot monitoring and availability information. Designed a distributed IoT architecture using ESP32 daughter nodes and a central gateway for sensor data collection. Built a FastAPI backend and React dashboard with dedicated interfaces for civilians, parking owners, and government authorities.

2. Fixed-Wing UAV – Unmanned Aerial Vehicle Development

Designed and developed a fixed-wing unmanned aerial vehicle as part of an aeromodelling and drone development competition. Worked on airframe assembly, aerodynamic structure, propulsion integration, and flight-control mechanisms.

3. Field-Eye – Autonomous Agricultural Rover

Developed an autonomous agricultural rover for real-time crop surveillance and early detection of pest-related abnormalities. Integrated ESP32-CAM, GPS, and wireless communication to capture and transmit field images and location data. Implemented AI-based image analysis to identify crop stress and pest presence, enabling timely alerts and targeted agricultural intervention.

4. SpinGen 2.0 – Kinetic Energy Harvesting System

Developed a hardware prototype to harvest wasted kinetic energy from rotating fans using an optimized magnet-coil electromagnetic induction mechanism. Designed a non-invasive energy harvesting system that converts mechanical motion into electrical energy.